Minqi (Christelle) Xiong

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Google Scholar | LinkedIn | GitHub | Personal Website

Research Interests

Personalized clinical decision support · Disease trajectory modeling · Digital twin · Learning health systems

Education

Johns Hopkins School of Medicine **Baltimore, MD**
M.S. in Health Sciences Informatics – Research *Aug. 2024 – Aug. 2026 (Expected)*
Certificate in Implementation Science and Research Practice

The Chinese University of Hong Kong **Shenzhen, China**
B.Sc. in Data Science & Big Data Technology *Sep. 2020 – May 2024*
Dean's List · First Prize, Netflix Recommendations Competition · First Prize, Kaggle COVID-19 Classification

Research Experience

Welch Center for Prevention, Epidemiology and Clinical Research **Baltimore, MD**
Graduate Researcher, PI: Dr. Anum Minhas *Feb. 2025 – Present*

Project *Cardiovascular Risk After Adverse Pregnancy Outcomes (APO)*
- Investigated the association between APO and subsequent cardiovascular disease risk using EHR data
- Developed computable phenotypes for pregnancy outcomes and cardiovascular outcomes using OHDSI ATLAS
- Co-authored a manuscript on cardiovascular risk after APO published in *Circulation*.

Project *Pregnancy Algorithm Evaluation*
- Evaluated OMOP-based pregnancy episode reconstruction algorithms against gold-standard registry data
- Examined agreement in gestational timing, pregnancy outcome classification, and complication capture across reconstructed pregnancy episodes and obstetric registry gold standards
- Preparing a first-author manuscript on pregnancy algorithm evaluation and maternal health outcome research
- Project repository: OMOP Pregnancy Algorithm

Decision Sciences Lab, Johns Hopkins School of Medicine **Baltimore, MD**
Graduate Researcher, PI: Dr. Harold P. Lehmann *Jan. 2025 – Present*

Project *Master's Thesis — Conceptual and Computational Framework of Disease Severity*
- Conducting a scoping review to identify how disease severity has been defined and conceptualized
- Developing a unified conceptual framework for multilevel and multidimensional disease severity representation
- Establishing a temporal modeling framework for characterizing disease severity trajectories

Project *Open-source Systematic & Scoping Review Platform*
- Developing an open-source platform for systematic and scoping reviews with integrated screening, corpus management, and reproducible evidence synthesis workflows
- Creating ontology-informed structures to organize disease concepts and evidence relationships
- Building automated workflow components to support scalable and reproducible review studies
- Project repository: Evidence Synthesis Platform

Center for Population Health Information Technology (CPHIT) **Baltimore, MD**
Graduate Researcher, PI: Dr. Hadi Kharrazi *Oct. 2024 – Present*

Project *Racial Disparities in Post-Hysterectomy Emergency Department Visits*
- Investigated racial disparities in 30-day ED utilization following hysterectomy using administrative data
- Constructed an analytic cohort of approximately 200,000 hysterectomy cases from 48M+ inpatient and outpatient records in the Healthcare Cost and Utilization Project (HCUP)

- Developed descriptive characteristics, multivariable logistic regression and sensitivity analysis workflows in R to evaluate disparities in post-surgical healthcare utilization
- Published a first-author peer-reviewed journal article in *International Journal of Gynecology and Obstetrics*

Project *Diagnostic Concordance Between Administrative Claims and Electronic Health Records*

- Investigated agreement and discrepancies between diagnoses recorded in administrative claims and electronic health records using Optum Labs data from 17M+ patients across 56 healthcare organizations
- Evaluated concordance and discrepancies between diagnosis codes recorded in claims and EHR systems
- Published a first-author peer-reviewed journal article in *JAMIA Open*

Myositis Center, Johns Hopkins Rheumatology **Baltimore, MD**
Graduate Researcher, PI: Dr. Christopher A. Mecoli *Oct. 2024 – Present*

Project *OHDSI Network Study of Dermatomyositis Adverse Events*

- Developed computable phenotypes and cohorts for dermatomyositis and immunosuppressants using the OMOP CDM and OHDSI ATLAS to support multi-institutional observational research
- Supported study design and comparative effectiveness analyses of immunosuppressant therapies and medication-related adverse events using large-scale OMOP-based observational data
- Contributed to the development of a Strategus R package for cohort discovery and evaluation
- Project repository: Guideline-driven Evidence in Rheumatology Infectious Study

Project *Steroid Dose Calculation and Exposure Profiling*

- Led the development of a framework for estimating corticosteroid daily dosage from EHR data
- Designed data pipelines to extract and normalize dosing information from medication prescriptions
- Implemented NLP and LLM-based methods for steroid dosage extraction from prescriptions and clinical notes
- Developed an open-source R package for reproducible steroid exposure calculation across EHR databases
- Preparing a first-author manuscript on steroid dose extraction and exposure profiling
- Project repository: Steroid Dosage Calculation

Human Language Technology Laboratory **Shenzhen, China**
Research Assistant, PI: Dr. Zhiheng Wu *Sep. 2022 – Jul. 2023*

Project *Accent and Speech Voice Conversion*

- Collected an audio data set from 40 volunteers and pre-processed and refined the 120 hours audio dataset
- Co-developed SoftVQ, a vector quantization framework for non-parallel many-to-many voice conversion
- Built an end-to-end deep learning pipeline integrating BN-PNet, SoftVQ modules, and autoregressive decoding
- Achieved improvements of 76.3 points in perceived audio quality and 1.1 points in speaker similarity in ABX evaluations using JavaScript-based questionnaires

Selected Publications

H-index: 4 † = *co-first authorship*

Peer-Reviewed Journal Articles

1. **Xiong C**, Wang Y, Dun C, Borahay M, Kharrazi H. Assessing the impact of racial disparities on 30-day emergency department visits due to hysterectomy complications: A multi-state retrospective cohort study. *Int J Gynaecol Obstet*.2026 Mar 13. doi: 10.1002/ijgo.70950. Epub ahead of print. PMID: 41822943. [**First & Corresponding Author**]

2. **Xiong C**, Pittell H, Kitchen C, Lasser E, Kharrazi H. Synergy of diagnosis coding between administrative claims and electronic health records of large patient populations across multiple healthcare organizations. *JAMIA Open*. 2026 Mar 11;9(2):ooag030. doi: 10.1093/jamiaopen/ooag030. PMID: 41836279; PMCID: PMC12986766..

3. Lee H, **Xiong C**, Baughman D, Dun C, Tong J, Martin B, Lehmann H, Nagy P. A multidimensional hierarchical framework for sources of bias in real-world healthcare evidence: a scoping review. *J Biomed Inform*. 2026 Feb;174:104989. doi:10.1016/j.jbi.2026.104989. Epub 2026 Jan 20. PMID: 41571173.

4. Boyer T, Barrett R, **Xiong C**, Bhansali R, Wallace A, Fang M, Vaught AJ, Selvin E, Ndumele C, Blaha M, Michos E,

Coresh J, Minhas A. Hypertensive Disorders of Pregnancy and Premature Cardiovascular Disease in a Diverse Cohort of Young US Women. *Circulation*. 2026 Feb 17;153(7):480-492. doi:10.1161/CIRCULATIONAHA.125.078057. Epub 2025 Nov 8. PMID: 41697979; PMCID: PMC12912777.

5. Chen D, Couperus CJ, Lee S, Barrett R, Lee H, **Xiong MC**, Wang Y, Wang Q, Lehmann HP. Evaluation of large language model performance in assessing health economic study quality. *J Health Econ Outcomes Res*. 2025 Oct 24;12(2):154-162. doi:10.36469/001c.145214. PMID: 41146963; PMCID:PMC12554303.

Manuscripts Under Review or In Preparation

6. Dobbins N[†], **Xiong C**[†], Lan K, Yetisgen M. LLM-based agents for automated research reproducibility in Alzheimer's disease. *J Med Internet Res*. **[Co-first Author; Under Review]**.

7. **Xiong C**, Lehmann HP. Conceptualizing disease severity: a scoping review and proposed conceptual framework. *J Biomed Inform*. **[In Preparation]**.

8. **Xiong C**, Barrett R, Boyer T, Martin B, Minhas A. Evaluation of pregnancy episode identification algorithms in real-world healthcare data. *JAMIA*. **[In Preparation]**.

9. Castner MD, Kitchen C, **Xiong C**, Bittle M, Nestadt PS, Wilcox HC, Kharrazi H. Assessing the utility of social determinants of health data in suicide prediction models. *Arch Suicide Res*. **[Under Review]**.

10. Castner MD, Kitchen C, **Xiong C**, Bittle M, Nestadt PS, Wilcox HC, Kharrazi H. Assessing the utility of social determinants of health and access data in ecological suicide prediction models. *Soc Psychiatry Psychiatr Epidemiol*. **[Under Review]**.

Conference Full Papers

13. **Xiong M**, Wu Z. Soft Vector Quantization for Non-Parallel Any-to-Many Voice Conversion. *Speech and Language Processing Technology Conference*. 2023. [Manuscript] [Poster]

14. Zhang H, **Xiong M**, Shi T, Liu W, Xu H, Zhao H, Kong G. Reinforcement Learning-based Decision-making for Renal Replacement Therapy in ICU-acquired AKI Patients. *ACM KDD 2024 Workshop on AI for Decision Support in Healthcare*. [Manuscript] [Poster]

Oral Presentations & Posters

O1.**Xiong C**, Wang Y, Dun C, Kharrazi H. Racial disparities in 30-day ED visits due to hysterectomy complications. *AMIA Annual Symposium 2025*. **[Podium; Selected for Recording & On-Demand Dissemination]** [Abstract] [Slides]

O2. **Xiong C**, Martin B, Westlund E, Kelly W, Mecoli C. AgentDose: Toward Accurate and Scalable Steroid Dose Extraction in OMOP Using NLP Parsers and LLM Agents. *OHDSI Global Symposium 2025*. **[Nominee, Best Community Contribution Showcase]** [Brief Report] [Poster]

Honors & Awards

2025: Selected Presentation, AMIA Annual Symposium (recorded presentation and on-demand dissemination)

2025: Best Community Contribution Showcase Nominee, OHDSI Global Symposium (AgentDose)

2025: Audience Favorite Award, Hexcite Healthcare Innovation Program, Johns Hopkins Technology Ventures

2023: Dean's List, The Chinese University of Hong Kong, Shenzhen

2023: First Prize, Netflix Recommendation Algorithm Competition

2023: First Prize, COVID-19 Behavioral Differences Classification Challenge (Kaggle)

2020: Bowen Scholarship (merit-based, $4,000/year)

Teaching & Mentoring

Johns Hopkins School of Medicine
Graduate Teaching Assistant *2025–2026*

Courses supported:
- ME.250.782 Observational Health Research Methods on Medical Records (OMOP)
- ME.250.861–864 Health Sciences Informatics Research Methods I–IV
- ME.250.788 Observational Research Methods in R

Self-Employed
Independent Academic Advisor *2021–Present*
Advised and mentored 27 high school and undergraduate students on undergraduate and graduate school applications, including CV preparation, personal statements, program selection, and long-term academic career planning

Professional Service

- Reviewer for journals: *Medical Care*, *JAMIA Open*
- Reviewer for conferences: AMIA Annual Symposium, Amplify Informatics Conference, Clinical Informatics Conference, IEEE International Conference on Health Informatics (ICHI)
- Member of American Medical Informatics Association (AMIA), Observational Health Data Science and Informatics (OHDSI), and American Psychological Association (APA)
- Program committe member of IEEE-ICHI 2026

Technical Skills

Programming	Python · R · SQL
ML Methods	Predictive modeling · survival analysis · causal inference · longitudinal modeling · natural language processing
ML Frameworks	PyTorch · Scikit-learn · XGBoost · LightGBM · spaCy
Healthcare Data	EHR · administrative claims · disease registries
Clinical Terminologies	ICD-10 · RxNorm · CPT · LOINC
Data Infrastructure	SQL Server · Google BigQuery · Databricks · Docker · Git
Software	VS Code · MATLAB · TreeAge · Overleaf

Additional Experience

Igor AI **United States**
Co-Founder *2025 – Present*
- Led the early-stage product design for an LLM-based platform supporting AI-assisted project management
- Contributed to discussions on system architecture and use cases for generative AI in collaborative platforms
- Exploring applications of LLMs to streamline task coordination and communication between team members and management

Hexcite: Excited for Healthcare, Johns Hopkins Technology Ventures **Baltimore, MD**
Technical Product Lead, GlioNet *Jan. 2025 – May 2025*
- Designed clinical UI and workflow integration for a precision medicine clinical decision support tool predicting glioblastoma survival and gene mutations from brain MRI and clinical features
- Contributed to product strategy, business model development, and product–market fit analysis
- Selected for **Audience Favorite Award** at the Hexcite Healthcare Innovation Program

EDGE Global AI & Web3 Summit, VertexLab **Hong Kong**
Conference Organizer & Interviewer *Apr. 2023 – Oct. 2023*
- Organized and moderated speaker panels for an international AI and Web3 technology summit
- Conducted live interviews with industry leaders on emerging AI technologies and startup ecosystems